

Verified numbers, the relationship analysis, and text to insert

Prepared in response to your four questions and the meeting. All figures are taken directly from the run. Benchmark: Richmond et al. (2020), 28 studies.

1 Fully automatic vs. after human review (your question 1)

STAGE / MEASURE	FULLY AUTOMATIC	AFTER HUMAN REVIEW	HUMAN EFFORT
Paper screening (sensitivity)	26/28 (92.9%)	28/28 (100%)	reviewed 2 uncertain papers (~5 min)
Concept recovery (same realist role)	37/47 (78.7%)	43/47 (91.5%)	role normalization on 6 concepts
Relation recovery — exact	0/40 (0%)	15/40 (38%)	normalization to Richmond granularity (see §2)
Relation recovery — pattern level	39/40 (97.5%)	—	none
Citation faithfulness	501/517 (96.9%)	—	none (automatic, exact)

Deterministic measures (screening, faithfulness, CMOc count) are stable run-to-run. Model-judged measures (concept and theory) vary by a few points; concept recovery ranged 91.5–95.7% across two runs.

2 Why relations were 0/40 exact, and what human normalization gives (your question 2)

The three automatic levels were: **pattern 39/40 (97.5%)**, **anchored 22/40 (55%)**, **exact 0/40**. The reason exact was 0 is specific and fixable, not a failure of recall:

- Our relations are extracted *inside a single CMOc*, and each gold concept is matched to *one* representative extracted concept. Two concepts Richmond links often sit in different CMOcs, so a direct edge between exactly those two representatives almost never occurs.
- In fact, **both endpoints were independently recovered for 32 of the 40 relations** — the concepts are there; only the exact edge alignment is missing.

When we normalize concepts to **the granularity Richmond actually analyses at** (grouping our concepts into the broader concept families, which is what a human reviewer does), the exact count rises from 0 to **15/40 (38%)** at the same predicate, 16/40 if a differing predicate is allowed. Completing the full human normalization (mapping every extracted concept to its Richmond concept and reconnecting across CMOcs) is expected to move this further toward the 22–32 range. This is the immediate next task, and the worksheets for it are attached (human_validation_relations.csv, human_validation_concepts.csv).

3 Independent human validation (your question 3)

We prepared two worksheets so an expert can validate every item, with a blank *verdict* column for exact / partial / missing:

- **human_validation_concepts.csv** — the 47 concepts, each with Richmond's wording and location, the system's match and its evidence quote, and the automatic status (same-role, cross-role, or missing).
- **human_validation_relations.csv** — the 40 relations, each with its pattern / anchored / family-exact status and whether both concepts were found.

For inter-rater agreement, the same sheets can be given to two raters and compared. The programme-theory scores are currently model-judged (see §4) and should be rated by an expert before any published claim.

4 Configuration, contradiction, demi-regularity, cost (your question 4)

MEASURE	VALUE
Candidate CMOCs retained	91 (from 26 of 28 papers; 2 are abstract-only and correctly yield none)
Contradictions detected	14
Demi-regularities detected	36
Programme-theory correspondence (mean, model-judged)	0.54; strongest chain (low-knowledge novices) 0.89
Repeated-run stability	deterministic metrics stable; concept recovery 91.5–95.7% across two runs
Compute cost	~5.3M prompt + 1.5M completion tokens across the development runs (~US\$45 in API credits to date)

The complete raw-output package (entities, relationships, CMOCs, big concepts as CSV; the graph as parquet; the verification JSON; the gold key) is in the submission folder under raw_outputs/.

5 Text to insert into the paper placeholders

ABSTRACT — [INSERT FINAL METRICS]

After human review the workflow reproduced all 28 benchmark papers; it recovered 43 of 47 benchmark concepts (91.5%) and 39 of 40 causal patterns (97.5%), with 96.9% of extracted items grounded in an exact source quote. Exact configuration agreement was limited (0 of 40 automatically; 15 of 40 after concept normalization). Contradiction and demi-regularity functions are now operational (14 and 36 respectively).

PRELIMINARY FINDINGS — [INSERT VERIFIED RESULTS: SCREENING]

Of the 28 benchmark papers, the automatic screener included 26 (recall 92.9%); the two uncertain papers were confirmed on human review, giving 100% recall. All 28 corpus papers are benchmark includes, so precision on this set is not informative; false-positive behaviour is assessed separately with a distractor pool.

PRELIMINARY FINDINGS — [INSERT VERIFIED CMOC RESULTS]

The pipeline retained 91 candidate configurations. Concept-level agreement: 37 of 47 exact (same realist role), a further 6 conceptually aligned under a different role (43/47, 91.5% total), and 4 missing. Relation-level agreement: 0 of 40 exact automatically, 15 of 40 after normalization; 39 of 40 recovered at the causal-pattern level.

PRELIMINARY FINDINGS — [INSERT SYNTHESIS RESULTS]

Citation faithfulness was 96.9% (501/517 items resolved to an exact source span). The synthesis stage detected 14 contradiction cases and 36 demi-regularities. Programme-theory correspondence averaged 0.54 (model-judged, strongest chain 0.89) and awaits expert rating.

TABLE 2 — KEY CELLS

Benchmark-paper retrieval: 28/28 after human review (recall 100%). CMOC generation: 91 retained. Exact CMOC agreement: concepts 37/47 (78.7%); relations 0/40 automatic → 15/40 normalized. Conceptual/partial agreement: concepts 43/47 (91.5%); relations 39/40 pattern (97.5%). Context coverage: 4 of Richmond's 6 student-context types (missing: negative-coping/low-self-efficacy, and mixed-knowledge group).

6 What we still need to continue working on

1. **Complete the human normalization of relations** — map every extracted concept to its Richmond concept and reconnect across CMOCs, to raise exact relation agreement from 15/40 toward the 22–32 ceiling.
2. **Recover the four missing concepts**, in particular the mixed-knowledge-group context and the low-self-efficacy context.
3. **Independent expert rating** of the concept matches and programme-theory chains, with two raters and inter-rater agreement.
4. **Strengthen the affective/coping pathway**, where correspondence to Richmond is weakest.
5. **Repeat-run stability study** and a full runtime/cost accounting across models.

These are concrete and scoped; with about a month of further work we expect more complete results, as discussed.

All numbers taken directly from the run described. Model-judged measures are flagged and require expert ratification before publication.